The Legitimacy Ledger

A conserved-quantity science of institutional trust: from the DPT impossibility theorem to metered governance, capture early warning, and the legitimacy-adjusted cost of capital.

Thesis — Legitimacy is not a perception to be surveyed; it is a conserved, meterable quantity that institutions accumulate or destroy one authorized transition at a time. The DPT theorem proves that every governed system must sacrifice exactly one of due process, partition tolerance, or autonomous termination — and the oscillation index Ω makes the sacrifice, and its abuse, measurable. This blueprint extends that result from machine systems to states, markets and development portfolios, converting governance from a compliance narrative into an instrumented, falsifiable asset class.

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C0 INFORMATIONAL · REGISTERED RESEARCH PROGRAMME — INSTRUMENTS FROZEN IN ADVANCE;
PROSECUTION-SENSITIVE DETAIL WITHHELD PENDING IP REVIEW

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How to read this edition

Parts I–III establish the problem and prove that it is unoccupied intellectual territory. Part IV states the formal extension of the parent theorem. Part V is the empirical programme — pre-registered, with explicit rejection criteria. Part VI specifies the instrument suite. Part VII states the risk register and the kill criteria the programme binds itself to. The Concluding Assessment compresses the argument for decision-makers. Every claim in this document is engineered to be held to account: where the evidence is thin, the document says so and proposes the test, not the conclusion.

CONTENTS

Table of Contents

Executive Summary

Part I — Strategic Context: The Governance Measurement Crisis

1.1 The paradox: the world prices everything except legitimacy
1.2 The perception ceiling: what indices cannot see
1.3 The capture wave: why the market forms now
1.4 The regulatory ratchet: auditability becomes law

Part II — The Parent Work: What the DPT Theorem Established

2.1 Traversals, not states: the ontological move
2.2 Conservation of legitimacy
2.3 The DPT impossibility theorem and its price list
2.4 The oscillation index and the enforcement schema

Part III — The Revolutionary Extension: Five Contributions

3.1 From theorem to accounts: the Legitimacy Accounting Standard
3.2 The trilemma as procurement law
3.3 Ω at sovereign scale: the capture early-warning instrument
3.4 Refusal-set economics and the LACC wedge
3.5 Compliance without transfer: ZK-attestation of authority

Part IV — Formal Framework: The Legitimacy Accounts

4.1 Stocks, flows and deficits
4.2 The trilemma frontier and pullback density
4.3 Formal specification
4.4 Constraint phenotypes
4.5 Four completed natural experiments

Part V — Empirical Programme and Methodology

5.1 Hypothesis portfolio
5.2 Corpora and access
5.3 Method, pre-registration and null models
5.4 Threats to validity and reflexivity discipline

Part VI — The Instrument Suite

6.1 LAS: the accounting schema
6.3 The Trilemma Audit protocol
6.5 The prospective pilots: two-site design

Part VII — Risk Register and Kill Criteria

Concluding Assessment

References

Appendix A — Glossary of Constructs

Appendix B — Indicator and Verification Matrix

Appendix C — Capture Case Files

Appendix D — Sensitivity Analysis

ORIENTATION

Executive Summary

The world has no meter for the thing it most needs to measure. Governments route roughly US$11 trillion a year — about 12 percent of global GDP — through public procurement;[18] the United Nations Secretary-General puts the annual cost of corruption at US$3.6 trillion in bribes and stolen money;[13] the World Bank committed US$40.9 billion in new IBRD lending in fiscal 2025 alone.[29] Yet the instruments used to judge the institutional quality on which all of this rides are perception surveys, published annually, which the World Bank itself warns are "estimates rather than definitive ratings" whose perceptions "may naturally lag behind actual policy reforms."[15] A balance sheet exists for every other asset class. Legitimacy — the property that makes a state's acts binding, a hospital's records trustworthy, and a procurement award defensible — is measured by asking people how they feel about it, once a year, with a lag.

The parent work supplies the missing physics. In *Legitimacy as a Conserved Quantity* (Odeta, 2026),[1] Dr. Odeta proved three results that this blueprint treats as foundational: that legitimacy is a property of *transitions*, not states, and is conserved under any operation preserving ledger integrity (Theorem 1); that no governed sync system can simultaneously guarantee due-processed conflict resolution, partition tolerance, and autonomous termination — the DPT impossibility theorem (Theorem 2), which partitions every design into three enumerable sacrifice regimes; and that constitutional capture emits a measurable advance signature, the oscillation index Ω, computable from amendment ledgers alone. A companion artifact enforces the theorem's boundary conditions at the database level, with seven falsifiability tests passing on a live instance.[1]

This blueprint carries the theorem from machines to institutions. It makes five contributions. First, the **Legitimacy Accounting Standard (LAS)**: a double-entry discipline that turns Theorem 1 into institutional accounts — a legitimacy balance sheet and flow statement for any governed body, metered from its own event ledger rather than from surveys. Second, the **Trilemma Audit**: DPT generalized into a procurement interrogation — every digital public infrastructure deployment must declare which of D, P, T it sacrifices; the vendor claim "automatic resolution of all conflicts" is now provably a claim of the impossible triple. Third, the **Governance Early-Warning Instrument (GEWI)**: Ω computed at sovereign scale from public amendment corpora — 200-plus national constitutions since 1789,[12] standards-body revision logs, and health-exchange incident records — validated retrodictively against the largest natural laboratory of capture ever recorded: the 44 countries currently in autocratization episodes, home to 41 percent of world population and 39 percent of world GDP.[25] Fourth, **refusal-set economics and the legitimacy-adjusted cost of capital (LACC)**: the financing premium attached to uninstrumented governance, and the option value of a stated refusal set — the system's declared Gödel sentence. Fifth, **ZK-attestation of authority**: compliance without data transfer, replacing the data-transfer agreement as the atomic unit of cross-border governance.[1]

The timing argument is fourfold. The capture wave is at a recorded peak — democracy for the average world citizen is back to 1978 levels.[25] The digital public infrastructure wave is deploying governed systems at national scale — MOSIP-based identity registered with more than 110 million people across the Philippines, Ethiopia and Morocco,[23] and DHIS2 serving as the government health-data system of record in more than 80 low- and middle-income countries[24] — each one a DPT deployment whether its designers know it or not. The regulatory wave has made auditability a legal obligation: the EU AI Act requires automatic event logging over the lifetime of high-risk AI

systems, with fines up to €15 million or 3 percent of global turnover, applicable from December 2027.[27],[28] And the measurement establishment has publicly conceded its own ceiling.[14],[15]

The programme's institutional form is deliberately conservative: an open accounting standard that creates the market, accompanied by four instruments (GEWI monitoring, Trilemma Audits, the Refusal Registry, ZK federation pilots) addressed to three constituencies — funders and multilateral development banks, states and authorities, and vendors and standards bodies. The empirical programme is pre-registered with explicit kill criteria: eight hypotheses, five corpora, permutation nulls, and a reflexivity protocol that forbids tuning the instrument to the result. If the programme confirms the framework, it owns the measurement layer of the governance economy. If it falsifies it, the rejection criteria will have taught the field exactly where the boundary lies — which is what falsifiable means.

PART I

Strategic Context: The Governance Measurement Crisis

1.1 The paradox: the world prices everything except legitimacy

Modern finance meters risk with obsessive granularity. Credit spreads move in basis points; catastrophe bonds price hurricanes; carbon markets price a ton of dioxide. Yet the single asset on which the entire development portfolio depends — the legitimacy of the institutions that receive, spend, and account for the money — has no meter, no unit, and no market. The exposure is not small. Public procurement alone runs at roughly US$11 trillion a year worldwide, about 12 percent of global GDP, with the share nearly identical across low-, middle- and high-income countries;[18] in most developing economies procurement exceeds 15 percent of GDP, and in Africa averages 17 percent.[20],[21] The World Bank committed US$40.9 billion in new IBRD lending across 139 operations in fiscal 2025, a 9 percent increase on the prior year.[29] Every dollar of it passes through institutions whose capacity to act legitimately — to procure, license, adjudicate, amend — is assessed, at best, once a year, by asking experts and households for their impressions.

The cost of getting this wrong is denominated in the trillions, though the honest literature refuses false precision. The UN Secretary-General cites US$3.6 trillion a year lost to bribes and stolen money;[13] the most-quoted estimate — US$2.6 trillion, 5 percent of global GDP — has been traced to advocacy documents rather than to any published methodology, and the World Bank's own governance blog now warns that all global corruption-cost statistics "rest on fragile foundations" and should be avoided in high-level use.[14] This concession is itself the strategic opening: the field's flagship numbers are unverifiable, its flagship indices are perceptual, and its flagship institution has said so in print. Table 1 assembles the verified contours of the crisis.

Table 1. The governance measurement crisis — verified indicators

Indicator	Value	Year	Source
Global public procurement	≈ US$11 tn / yr (12% of GDP)	2018	World Bank[18]
Procurement share of GDP, developing economies	15–22%	2020	World Bank[19]
Procurement share of GDP, Africa (average)	17%	2023	World Bank[20]
Procurement share of GDP, OECD	12.7%	2023	OECD[21]
Estimated annual cost of corruption (bribes + stolen funds)	US$3.6 tn	2018	UN Secretary-General[13]
Status of global corruption-cost estimates	"rest on fragile foundations"	2022	World Bank[14]
WGI design basis	perception only; 35 sources; 200+ economies; annual	2025	World Bank[15],[16]
WGI usage advisory	"estimates rather than definitive ratings… perceptions may naturally lag behind actual policy reforms"	2026	World Bank[15]
Countries in autocratization episodes	44 (record)	2025	V-Dem[25]

Population living in autocratizing countries	3.4 bn (41% of world)	2025	V-Dem[25]
Share of world GDP in autocratizing countries	39%	2025	V-Dem[25]
Episode "fatality rate" (democracies that began autocratizing and remained democracies)	9 of 27 survived	2024	V-Dem[26]
Contracts openly published	2.8% of spend	2021	World Bank[22]
IBRD net commitments	US$40.9 bn (139 operations)	FY2025	IBRD financial statements[29]
DHIS2 deployments (government health system of record)	80+ LMICs; 100+ incl. NGOs	2026	DHIS2/HISP UiO[24]
MOSIP-based national IDs issued	110 m+ (PH, ET, MA)	2024	MOSIP ecosystem[23]
EU AI Act record-keeping duty (Art. 12) — penalties	up to €15 m or 3% turnover	from Dec 2027	Reg. 2024/1689; 2026/1744[27],[28]

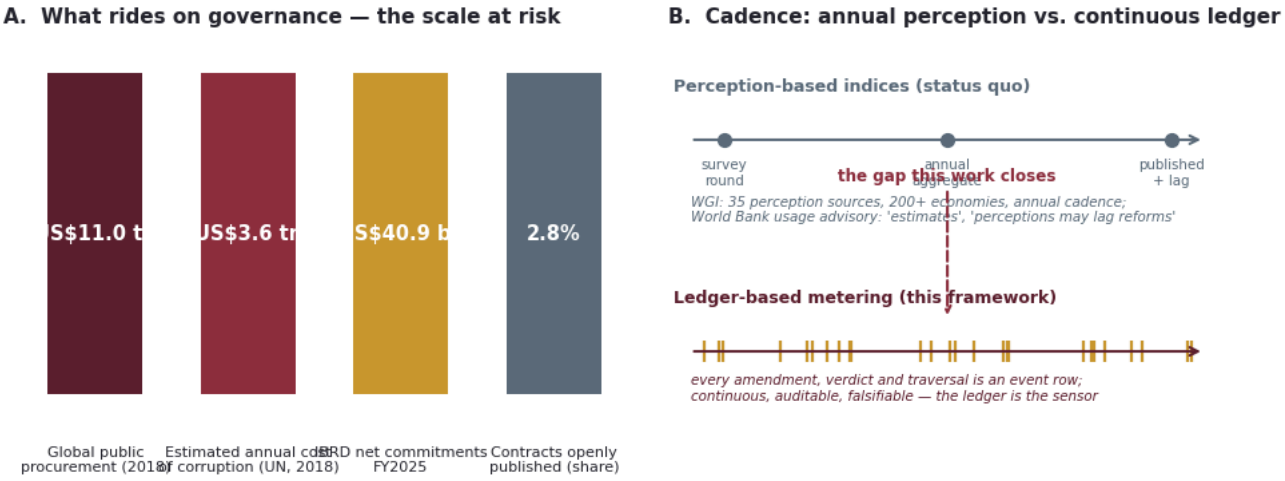


Figure 1. The measurement gap. Panel A: verified magnitudes riding on institutional legitimacy — global procurement,[18] the UN corruption-cost estimate,[13] FY2025 IBRD commitments,[29] and the 2.8 percent of contract spend that is openly published.[22] Panel B: the cadence asymmetry this framework closes — annual perception aggregates versus a continuous event ledger in which every amendment, verdict and traversal is an instrumented row.

1.2 The perception ceiling: what indices cannot see

The Worldwide Governance Indicators are the field's workhorse: six dimensions, more than 200 economies, built exclusively from perception sources — household and firm surveys plus expert assessments.[16] Their architects are candid about the design: the WGI methodology document states that the framework "is perception-based," and the current usage advisory instructs that scores "should be treated as estimates rather than definitive ratings," that they "should not serve as definitive criteria for credit ratings, investment risk, or other critical financial decisions," and that "public perceptions may naturally lag behind actual policy reforms."[15] The academic literature adds three structural criticisms: aggregation across heterogeneous perception sources introduces validity and reliability problems; country-level composites erase sub-national variation; and the underlying construct embeds normative assumptions about what good governance is.[17] A

long-standing operational critique is non-actionability: an indicator can report that businesspeople feel property rights are insecure, but not *why* — which rule changed, which organ changed it, and whether the change followed its own procedure.[17]

The ceiling is therefore not a failure of effort but a category error. Perception instruments measure what legitimacy *feels like* from outside; they cannot measure what legitimacy *is* from inside — a property of transitions, accumulated or destroyed one authorized act at a time. Capture, the phenomenon funders most need to anticipate, is precisely the phenomenon perception sees last: it announces itself first as procedural acceleration — amendments that invalidate prior-version acts arriving faster, cooling-off discipline eroding — long before any substantive failure is visible to survey respondents.[1] The sensor for that signature already exists in every governed system. It is the amendment ledger. Nobody is reading it as an instrument.

1.3 The capture wave: why the market forms now

The demand side is a documented emergency. V-Dem's Democracy Report 2026 records 44 countries in ongoing autocratization episodes — the highest count ever registered — home to 3.4 billion people, 41 percent of world population, and generating 39 percent of world GDP; the average world citizen's democracy level is back to 1978, erasing nearly all gains of the third wave of democratization.[25] The epidemiology is unfavorable: of the 27 autocratization episodes that began in democracies tracked through 2024, only 9 remained democracies — a two-thirds fatality rate.[26] The autocrat's most common instrument is procedural: media censorship leads (32 countries, 73 percent of autocratizers), followed by civil-society repression — rule-change deployed faster than review can contest it.[25] That is, in the parent work's vocabulary, capture presenting as oscillation: high amendment velocity with falling cooling-off compliance. The phenomenon V-Dem catalogs at country level is the same signature the parent work formalized at ledger level. What has never existed is the bridge: an instrument that reads the procedural signature *before* the outcome data arrive.

Simultaneously, the supply side of governed systems is exploding. Digital public infrastructure is being deployed at national scale across the developing world: MOSIP-based foundational ID systems had registered more than 110 million people by early 2024, with full implementations in the Philippines, Ethiopia and Morocco and pilots or preparation in at least eight more countries;[23] DHIS2 is the government-wide health-data system of record in more than 80 low- and middle-income countries, explicitly designed for limited-connectivity operation;[24] and the Global Digital Compact — the first comprehensive multilateral framework on digital governance, adopted in September 2024 — had drawn 315 endorsements from entities in 93 countries by end-2024.[30] Every one of these deployments is an offline-tolerant, conflict-prone, authority-gated sync system. Every one of them, whether its designers know it or not, has already chosen a DPT sacrifice regime — almost never consciously, almost never auditably.

1.4 The regulatory ratchet: auditability becomes law

The third timing leg is legal. The EU AI Act's Article 12 requires that high-risk AI systems "technically allow for the automatic recording of events (logs) over the lifetime of the system," sufficient for risk identification, post-market monitoring, and operational oversight; deployers must retain logs at least six months, market surveillance authorities must be able to examine them on request, and violations carry fines up to €15 million or 3 percent of global annual turnover.[27] The 2026 Digital Omnibus (Regulation (EU) 2026/1744) moved the high-risk obligations to 2 December 2027 — a postponement that compliance analysts read as a build window, not a reprieve, because audit-grade evidence requires a year or more of production history to accumulate.[28] Critically,

commentators note what the regulation does *not* specify: how to protect logs from tampering after the fact — "a standard database record can be altered without trace," and only cryptographic chaining closes the gap between technical compliance and legal evidence.[28] That gap — automatic logging mandated by law, integrity left to the implementer — is exactly the gap the parent work's hash-chained amendment ledger and verdict lexicon already close in executable form.[1] Regulation has created a compliance market for the parent work's core mechanism before the market knows the mechanism exists.

Synthesis: a trillion-dollar exposure class measured by annual perception; a record capture wave whose early signature is procedural and therefore ledger-visible; a deployment wave of governed systems choosing sacrifice regimes unconsciously; and a regulatory ratchet mandating the parent work's primitive. Four independent trends, one vacant intersection. That intersection is the subject of this blueprint.

PART II

The Parent Work: What the DPT Theorem Established

2.1 Traversals, not states: the ontological move

The parent work's first move is ontological, and everything else inherits from it. Distributed systems theory knows how to make state converge under partition but attaches no normative weight to the process; constitutional law knows that authority must be bounded, revocable and contestable but cannot meter legitimacy; health informatics knows how to represent clinical semantics but treats governance as deployment paperwork.[1] The paper's response is to change the object of analysis: a governed system "does not converge toward states; it executes traversals" — six-phase authorization loops (observe, infer, authorize, act, verify, return), each phase gated by an authority function computed under the constitutional version in force at that moment.[1] Authority is computed, never possessed. Correctness is a property of states; legitimacy is a property of transitions. This single reframe is what makes the quantity meterable: states are unobservable in aggregate, but transitions are events, and events can be logged.

2.2 Conservation of legitimacy

Definition 3 of the parent work makes a transition legitimate iff three clauses hold jointly: (i) authority at traversal time exceeded zero under the constitutional version then in force; (ii) the transition is recorded in an append-only hash-chained ledger whose integrity is verifiable independently of payload accessibility; and (iii) the transition remains contestable at subsequent traversals.[1] Legitimacy is then an eigenform — the fixed point of the verification operation: re-running the constitutional traversal on a transition's provenance returns the same verdict. Theorem 1 follows: for any governed system with an unforgeable append-only ledger, net legitimacy — the sum over transitions of $\ell(\delta) \in \{+1, -1\}$ — is conserved under all operations preserving chain integrity, and strictly decreases under any ultra-vires transition; no subsequent operation can restore it without an adversarial ledger rewrite, which the Haber–Stornetta hash chain makes computationally visible.[1],[7] The proof sketch carries the deep point: correctness of content can be revised by later information, but legitimacy cannot, because clause (i) is indexed to the version in force at the moment of traversal. A later, better-informed traversal does not repair an illegitimate predecessor; it adds a new, separately legitimate transition. Illegitimacy, once incurred, is a permanent liability on the institutional balance sheet — visible forever.

2.3 The DPT impossibility theorem and its price list

The paper's central result — Theorem 2 — defines three properties for a governed sync system: **D** (due-processed resolution: every conflict resolves only through a procedure in which both branches' authority chains are presented, contestable, and the verdict cites its merge-ordering criteria); **P** (partition tolerance: traversals continue on each side of a network partition without halt); and **T** (autonomous termination: every conflict reaches a verdict without human traversal). It then proves no system guarantees all three. The proof's engine is causal: under a partition, a consent-withdrawal on one side is causally invisible to the other; the contestant "does not exist" in the resolving side's causal past; contestation is impossible in principle, not merely unimplemented — which is why no engineering improvement dissolves the theorem and only renegotiating the property set does.[1] DPT thereby joins the CAP family of distributed-systems impossibility results[4],[5] but differs in kind: the sacrificed property is procedural, not a data-consistency property, and the adversary is the FLP asynchronous scheduler[6] realized as jurisdictional partition.

The theorem's economic content is its price list. Every deployment sacrifices exactly one vertex, and the parent work tabulates the regimes: drop D and conflicts resolve silently while the legitimacy ledger accrues negative quantity invisibly; drop P and nodes halt under partition — field clinics go dark in the 18-hours-a-day offline reality; drop T (partially) and conflicts queue for human adjudication under explicit merge-ordering rules.[1] The CLAI-OS clinical architecture had selected the third regime *before* the theorem was stated — convergent validation that the framework describes real design pressure rather than imposing an abstraction.[1] The boundary conditions matter as much as the result: DPT governs only conflicts with empty policy pullback; independent additions and non-overlapping fields have non-empty pullback and are exempt. "The theorem partitions the operation space; it does not paralyze it."[1]

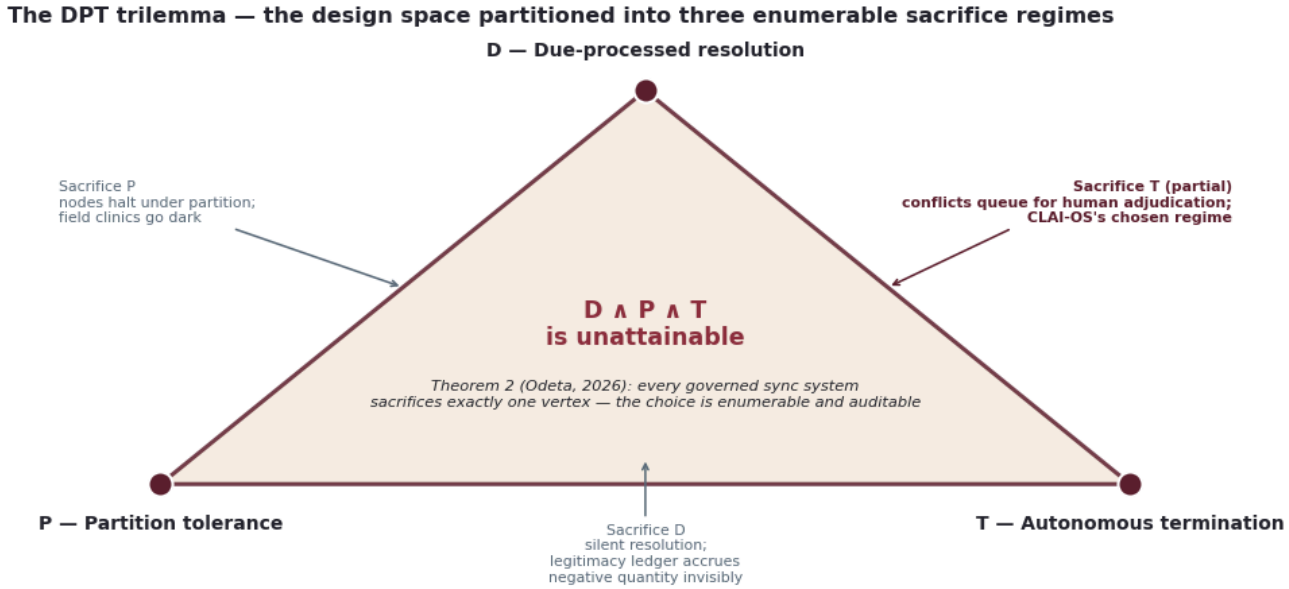


Figure 2. The DPT trilemma. Theorem 2 of the parent work partitions every governed sync design into three enumerable sacrifice regimes.[1] The procurement consequence follows immediately: a vendor claiming automatic resolution of all conflicts is claiming the impossible triple, and the question *which of D, P, T do you sacrifice?* is well-posed and unanswerable by marketing.

2.4 The oscillation index and the enforcement schema

The parent work's third pillar turns capture into telemetry. Over a ledger, define version-flip frequency φ(t) — amendments per unit time whose effect invalidates acts authorized under the immediately previous version — and cooling-off compliance κ(t) ∈ [0,1] — the fraction of amendments respecting their tier's mandatory delay. The oscillation index Ω(t) = φ(t)·(1−κ(t)) is proposed, as a falsifiable proposition, to rise monotonically in a window preceding visible failure in systems approaching capture, and to remain bounded by a low constant in healthy ones — requiring no telemetry beyond the amendment ledger itself.[1] The enforcement companion (Migration 1.1.0) writes the boundary conditions into a live schema: a closed verdict lexicon including NON_JUSTICIABLE — the system's declared Gödel sentence, distinguishing refusal to verdict from a negative verdict; structural amendment friction enforcing 7/90/365-day cooling-off tiers, mandatory sunsets and entrenchment rejection at INSERT, making capture "not prohibited by policy but unrepresentable in the data model"; and a hash-chained migration ledger on which seven falsifiability tests pass against a live instance.[1] This document treats all three pillars as established parent results and builds the superstructure of Parts III–VI on top of them.

PART III

The Revolutionary Extension: Five Contributions

The parent work proved its results for governed machine systems and pointed, in its discussion, at procurement, regulation and standards as the practices that would change if the programme held.[1] This Part makes the five moves that carry the framework from machines to institutions — and from a paper to an asset class. Each is stated as a thesis with its dependency on the parent results made explicit.

3.1 From theorem to accounts: the Legitimacy Accounting Standard

Theorem 1 establishes that legitimacy is conserved and that ultra-vires transitions permanently reduce it.[1] A conserved quantity that strictly decreases under identifiable events is, by definition, an accounting quantity: it supports a stock, a flow, and an audit trail. The first contribution is therefore to do for institutions what double-entry bookkeeping did for firms — define the **Legitimacy Accounting Standard (LAS)**: a chart of legitimacy accounts in which every governed transition δ is posted with its score $\ell(\delta)$, the constitutional version $\nu(t)$ under which it was authorized, the organ that executed it, and its contestation status. From the postings fall three statements: the **legitimacy balance sheet** (stock L(t), accumulated ultra-vires deficit U(t), and the capture-signature telemetry $\Omega(t)$); the **legitimacy flow statement** (legitimate transitions per period, decomposed by tier and organ); and the **contestation ledger** (the pipeline of challenged transitions and their dispositions). The standard's radical property is its data source: it is metered from the institution's own event ledger, not from surveys. Where the WGI aggregates 35 external perception feeds into an annual estimate,[15],[16] the LAS reads the instrument the institution already produces every day — and, per Theorem 1, cannot falsify retroactively without visible ledger surgery.

3.2 The trilemma as procurement law

The second contribution generalizes Theorem 2 into a procurement instrument. Theorem 2's proof is domain-general: its adversary is jurisdictional partition — the normal operating condition of national digital infrastructure, where connectivity gaps, cross-border jurisdictions and offline field operations partition the system daily.[1] Every digital public infrastructure deployment — identity registries, payment switches, health exchanges, land registries — is therefore a DPT system, and every such deployment has already sacrificed exactly one of D, P, T, whether or not anyone decided which. The **Trilemma Audit** is the interrogation protocol that forces the choice into the open at procurement time: (i) map the system's conflict classes and compute which have empty policy pullback; (ii) for each empty-pullback class, extract the deployed resolution path and identify the sacrificed vertex; (iii) score the sacrifice for explicitness, contestability and reversibility; (iv) write the result into the procurement file as a declared regime. The protocol converts the parent work's closing challenge — "a vendor claiming automatic resolution of all conflicts is claiming the impossible triple, and the question which of D, P, T do you sacrifice? is now well-posed and unanswerable by marketing"[1] — into a standard clause for the US$11 trillion procurement market.[18]

3.3 Ω at sovereign scale: the capture early-warning instrument

The third contribution scales Proposition 1 from machine ledgers to states. National constitutions, statutory frameworks and regulatory rulebooks are amendment ledgers: versioned rule histories with timestamps, tier classifications and procedural-delay requirements — precisely the data

structure Ω consumes. The Comparative Constitutions Project alone supplies amendment histories for more than 200 national constitutions from 1789 to the present;[12] official gazettes and parliamentary records extend the same structure to statutory and regulatory tiers; standards bodies (IETF obsolescence chains, HL7 FHIR release churn) supply non-sovereign control corpora.[1] The **Governance Early-Warning Instrument (GEWI)** computes φ, κ and Ω on rolling windows over these public corpora and tests the lead time Δ between Ω excursions and independently documented failure events against a permutation null. The validation laboratory is historical fact: 44 countries are currently in autocratization episodes,[25] and the standard tactic — procedural acceleration, rule-change outpacing review — is the signature Ω is designed to detect.[1] If the retrodictive programme confirms the lead time, funders acquire what they have never had: a continuous, cheap, event-based leading indicator of institutional capture, computable for every member country from data that already exists.

GEWI's early-warning reading is reconciled with the prediction/authority distinction in Appendix H-H (p. 33).

3.4 Refusal-set economics and the LACC wedge

The fourth contribution prices the framework. The parent work introduced NON_JUSTICIABLE — the declared refusal to produce a verdict, distinct from a negative verdict — as the system's stated Gödel sentence.[1] The economic reading is that a **stated refusal set is an asset**: it bounds the institution's discretion ex ante, converts silent failure modes into declared ones, and — per the parent work's hypothesis H4 — should clear regulatory review faster than functionally identical systems without it.[1] The mirror image is a liability: uninstrumented governance — institutions whose transitions cannot be audited, whose amendment discipline cannot be read, whose refusals are undeclared — imposes verification costs on every counterparty. Capital markets already price such opacity informally; this blueprint names the construct: the **legitimacy-adjusted cost of capital (LACC)**, in which auditability A and trailing capture-signature exposure Ω enter the required return alongside conventional risk premia. The testable implication is a wedge: two institutions identical on macro fundamentals but different on (A, Ω) should face systematically different financing terms. Measuring that wedge — and then compressing it by instrumenting the institution — is the framework's revenue logic in one line: **legitimacy, once metered, becomes bankable**.

3.5 Compliance without transfer: ZK-attestation of authority

The fifth contribution adopts the parent work's stated future work as a flagship programme. Cross-border governance currently runs on data-transfer agreements: to verify that an act was authorized, the verifier demands the evidence state — patient records, identity attributes, registry entries — under a negotiated transfer regime. The parent work specifies the alternative: zero-knowledge attestation of authority — proving that authority exceeded zero under policy version ν without revealing the evidence state E.[1] The atomic unit of cross-border governance becomes the attestation, not the transfer. The timing is precise: health data exchanges are federating under sovereignty constraints (MyHealth@EU-class architectures), national ID systems are scaling past 110 million registrations on sovereignty-sensitive terms,[23] and the Global Digital Compact has made data governance a first-order multilateral dossier.[30] A working ZK-attestation protocol converts the deepest structural conflict in digital development — verification versus sovereignty — from a negotiating deadlock into an engineering specification.

Table 2. The five contributions — dependency, novelty and falsification handle

Contribution	Builds on	Novel claim	Falsification handle
LAS — legitimacy			

accounts	Theorem 1; Def. 3; hash-chained ledger[1]	Legitimacy supports double-entry institutional accounting metered from event ledgers	LAS statements fail to reconcile on a live governed system
Trilemma Audit	Theorem 2; price list[1]	Every DPI deployment has an identifiable, auditable sacrifice regime	A deployed system exhibiting D ∧ P ∧ T on empty-pullback conflicts
GEWI — sovereign Ω	Def. 6; Proposition 1[1]	Ω computed from public amendment corpora leads documented failure events	No lead time in ≥50% of retrodictive cases (H1 rejection)
LACC wedge	Refusal-set semantics; H4[1]	Auditability and Ω price into financing terms	No significant wedge controlling for macro fundamentals (H7 rejection)
ZK attestation	Authority function; future work (iii)[1]	Compliance can be proven without evidence transfer at production scale	Proof cost/latency exceeds transfer-agreement cost in pilot (H8 rejection)

PART IV

Formal Framework: The Legitimacy Accounts

4.1 Stocks, flows and deficits

Let an institution be represented, per the parent work, as a traversal-executing system over a hash-chained ledger Λ, with constitutional versions ν(t) and transition legitimacy ℓ(δ) ∈ {+1, −1} per Definition 3.[1] The extension introduces the accounting objects. The **legitimacy stock** L(t) is the cumulative net score of recorded transitions; the **ultra-vires deficit** U(t) is the cumulative count of transitions failing clause (i) of Definition 3; and the **legitimacy flow** Φ(t) is the rate of legitimate transitions per period, decomposable by amendment tier and executing organ. Conservation (Theorem 1) becomes the accounting identity — relation (1) below — whose right-hand side is computable from Λ alone. The identity has teeth: any reconciliation failure between independently maintained copies of the ledger is, by the hash chain, computationally visible evidence of tampering rather than a data-quality footnote.[7]

4.2 The trilemma frontier and pullback density

For a deployed system S with conflict classes C, let ρ_PB(S) be the **pullback density**: the share of conflict classes whose governing policy pairs have non-empty pullback — where, per the parent work, the pushout exists and the class is exempt from Theorem 2.[1],[9] The theorem then bounds the attainable property set on the residual (empty-pullback) classes: at most two of D, P, T can hold simultaneously — relation (3) — and the audit problem is to identify which two. Pullback density also carries the framework's structural prediction: failures should cluster where pullback is empty (hypothesis H2 of the parent work[1]), because those are exactly the classes where the system is forced onto the trilemma frontier and any undeclared sacrifice becomes a latent legitimacy liability.

4.3 Formal specification

Seven relations define the extended framework. Quantities are per institution i and period t unless noted; parent-work constructs (Λ, ν, ℓ, φ, κ, Ω) are imported unchanged.[1]

(1) The conservation identity (accounting form). For institution i with ledger Λ_i:

$$L_i(t) = L_i(0) + \sum_{\delta \in \Lambda_i,\, \tau(\delta) \le t} \ell(\delta) \;-\; U_i(t)$$

where $U_i(t)$ counts transitions recorded with $\ell = -1$ (ultra-vires at traversal time). By Theorem 1, no sequence of subsequent legitimate transitions restores L: repair is additive, never retroactive.[1]

(2) The oscillation index (sovereign form). With tier-weighted version-flip frequency and cooling-off compliance over window w:

$$\Omega_i(t;w) = \varphi_i(t;w) \cdot \big(1 - \kappa_i(t;w)\big), \qquad \varphi_i = \sum_{k \in \text{tiers}} \omega_k \, \phi_i^{(k)}$$

Constitutional-tier flips (ω_k highest) invalidate the largest stock of prior-version acts; statutory and regulatory tiers scale down. The parent work's Proposition 1 — Ω rises before visible failure — is retained verbatim as the instrument's core falsifiable claim.[1]

(3) The trilemma inequality. For the empty-pullback conflict classes $C^{\varnothing}$ of system S:

$$1[D] + 1[P] + 1[T] \;\leq\; 2 \quad \text{on } C^{\varnothing}(S)$$

with equality universal in deployment: every fielded system saturates the bound, and the audit's task is to identify which indicator is zero. The regime is a choice; making it a *declared* choice is the Trilemma Audit's entire demand.

(4) Pullback density and the failure-clustering prediction.

$$\rho_{PB}(S) = \frac{|\{\, c \in C(S) : p_1 \times_a p_2 \neq \varnothing \,\}|}{|C(S)|}, \qquad \Pr[\text{failure} \mid c] \text{ decreasing in } \rho_{PB}$$

The second clause is hypothesis H2 in structural form: failure events concentrate where the pushout does not exist.[1]

(5) The legitimacy-adjusted cost of capital.

$$r_i = r_f + \rho\,(1 - A_i) + \pi \cdot \bar{\Omega}_i \cdot (1 - A_i)^{1/2}$$

where $A_i \in [0,1]$ is auditability — the share of governed transitions posted to a verifiable ledger — and Ω_i is the trailing capture-signature exposure. Uninstrumented governance ($A \to 0$) pays both premia in full; full instrumentation with a bounded signature drives both toward zero. The wedge between instrumented and uninstrumented twins is hypothesis H7.

(6) The early-warning lead time. Let t_Ω be the first crossing of a pre-registered Ω threshold and t_F the independently documented failure event:

$$\Delta = t_F - t_\Omega, \qquad H_1 :\; \mathbb{E}[\Delta] > 0 \text{ vs. permutation null (10,000 shuffles)}$$

The test protocol — rolling windows, no tuning to outcomes, pre-registered thresholds — is inherited from the parent work's retrodictive design.[1]

(7) The value of auditability. The instrument's economic content, per deployment:

$$V_A = \underbrace{\Delta_{\text{reg}} \cdot c_d}_{\text{review days saved}} + \underbrace{(r_i^{uninst} - r_i^{inst}) \cdot K_i}_{\text{financing wedge}} + \underbrace{p_F \cdot C_F \cdot (1 - A_i)}_{\text{expected failure cost avoided}}$$

Regulatory-review days (H4), the financing wedge (H7), and avoided silent-corruption failures (H3) are the three measurable channels through which metering pays for itself.

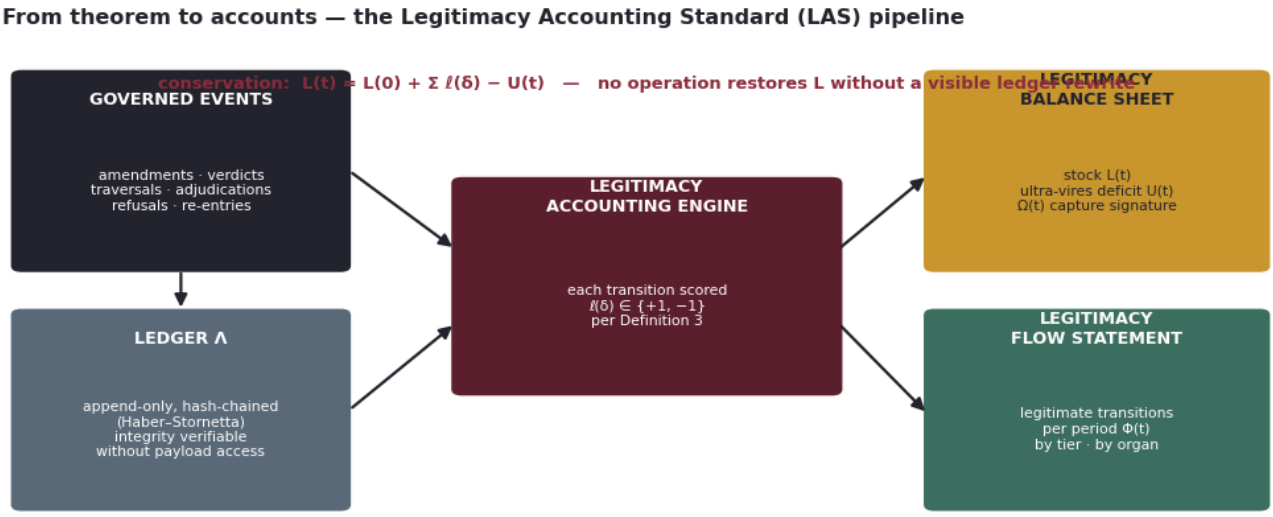


Figure 3. From theorem to accounts. The LAS pipeline: governed events post to the append-only hash-chained ledger Λ;[1],[7] the accounting engine scores each transition per Definition 3;[1] the outputs are a legitimacy balance sheet (stock L, deficit U, signature Ω) and a flow statement (Φ by tier and organ). Conservation — L(t) = L(0) + Σℓ(δ) − U(t) — is Theorem 1 in bookkeeping form.

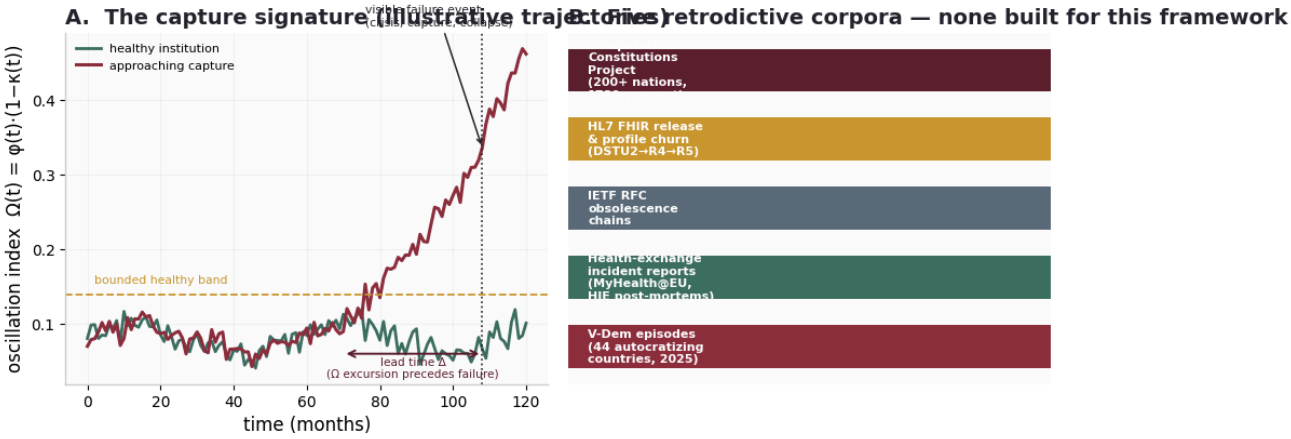


Figure 4. The capture signature and its test bed. Panel A (illustrative trajectories): healthy systems hold Ω inside a bounded band; systems approaching capture show rising version-flip frequency with eroding cooling-off compliance, crossing threshold Δ months before the visible failure event — Proposition 1 of the parent work.[1] Panel B: the five retrodictive corpora — constitutional amendment histories,[12] FHIR release churn,[11] IETF obsolescence chains,[1] health-exchange incident reports,[1] and V-Dem episode data.[25] None was built with this framework in mind; that is the point.

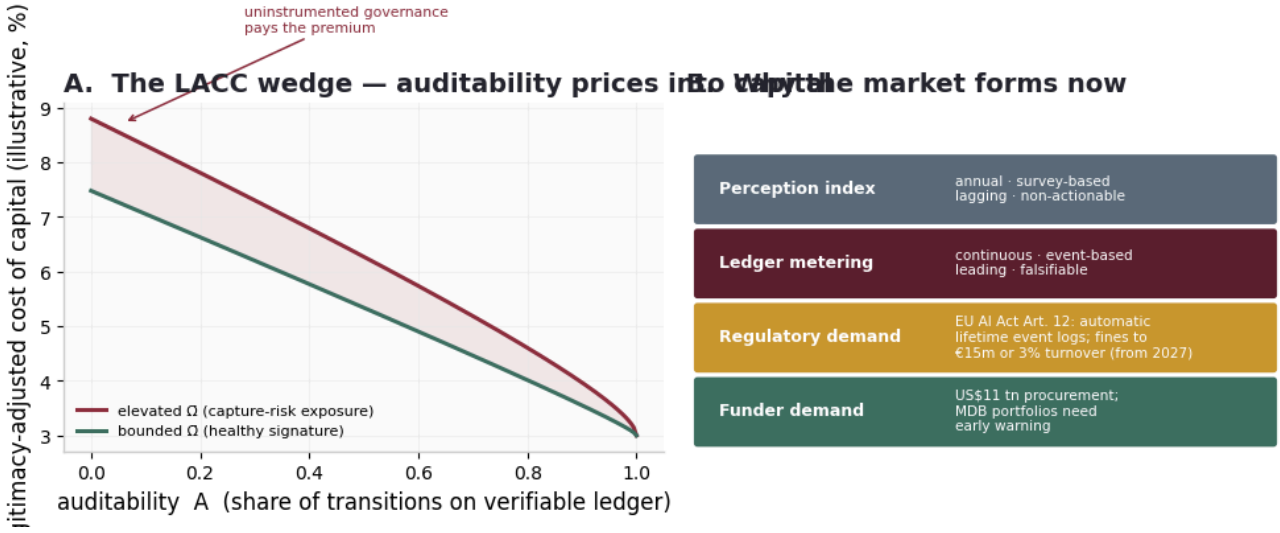


Figure 5. The LACC wedge (illustrative). Panel A: relation (5) plotted for two trailing-signature regimes — auditability compresses the legitimacy premium in both, but the wedge against an elevated-Ω twin is the financing argument for instrumentation. Panel B: the four forces converging on the market — perception indices' conceded ceiling,[15] ledger metering, the EU AI Act's Article 12 logging mandate,[27] and funder exposure across US$11 trillion of procurement.[18]

4.4 Constraint phenotypes

Five failure phenotypes recur across governed systems, each mapping to a construct of the framework and a primary instrument response. The phenotype table is the diagnostic front end of every engagement: classify first, instrument second.

Table 3. Constraint phenotypes — signatures, evidence and instrument response

Phenotype	Signature	Diagnostic evidence	Primary instrument response
G1 — Silent-resolution	Conflicts merged without presented authority chains; legitimacy ledger accrues invisible deficit	D dropped: no adjudication trail; U(t) rising while L appears stable	Trilemma Audit; verdict lexicon with cited merge-ordering criteria[1]
G2 — Partition-blind	Nodes halt or fork under partition; field operations dark	P dropped: traversals suspended during connectivity gaps[24]	Regime declaration; offline-first traversal design with deferred adjudication
G3 — Termination-blind	Conflicts queue indefinitely; adjudication backlog becomes shadow policy	T dropped without queue discipline: verdict latency unbounded	Merge-ordering rules with deterministic tie-break; queue telemetry[1]
G4 — Capture-approaching	Amendment velocity rising; cooling-off compliance eroding; prior-version acts invalidated at speed	Ω excursion above pre-registered threshold; κ(t) falling across tiers	GEWI alert; entrenchment friction; sunset enforcement[1]
G5 — Refusal-absent	System never declines to verdict; re-entries unrecorded; discretion unbounded	No NON_JUSTICIABLE-class verdicts in lexicon; constitution judging its own amendment silently[1]	Refusal Registry; stated refusal set; human-traversal routing for re-entries

4.5 Four completed natural experiments

The framework's components have each already been stress-tested by events none of its authors controlled.

Table 4. Natural experiments E1–E4

Experiment	Event	Framework reading
E1 — the convergence shock	CLAI-OS selected partial-T sacrifice (merge-ordering rules plus adjudication queue) *before* Theorem 2 was stated[1]	Convergent validation: real design pressure discovered the theorem's correct regime independently — the price list describes reality, not convention
E2 — the enforcement shock	Migration 1.1.0 wrote cooling-off tiers, sunsets, entrenchment and the verdict lexicon into a live SQLite schema; seven falsifiability tests pass, refused by the database engine, not application code[1]	The theorem's boundary conditions are executable at negligible cost — auditability is an engineering choice, not a budget line
E3 — the regulatory shock	EU AI Act Art. 12 mandates automatic lifetime event logging for high-risk systems from December 2027, with fines to €15 m / 3% of turnover — while leaving log integrity unspecified[27],[28]	Law has mandated the parent work's primitive (automatic event ledgers) and left open exactly the gap its hash-chaining closes — demand arrives pre-built
E4 — the capture shock	44 countries in concurrent autocratization episodes; dominant tactic is procedural (media censorship 73%, civil-society repression 68%) [25]	The largest-ever synchronized sample of the phenomenon Ω is designed to detect — a retrodictive laboratory at sovereign scale

PART V

Empirical Programme and Methodology

5.1 Hypothesis portfolio

The programme inherits hypotheses H1–H4 from the parent work verbatim — with their rejection criteria — and adds H5–H8 for the sovereign and financial extensions.[1] Every hypothesis carries a pre-registered rejection criterion; the programme is designed to be capable of failing.

Table 5. Hypothesis register H1–H8 with pre-registered rejection criteria

ID	Hypothesis	Rejection criterion
H1	Ω rises before documented governance failures (constitutional crises, standards-body capture, interoperability collapse)[1]	No lead time in ≥50% of failure cases
H2	Failure events cluster where participating jurisdictions' policies have empty pullback over the exchanged data class[1]	Failures distribute uniformly across pullback density
H3	Failures present as silent semantic corruption — datum valid in each system individually, invalid jointly — not as rejected transfers[1]	Failures present predominantly as plain rejection or transport error
H4	Deployments with complete sealed-payload auditability clear regulatory review (DPIA-class) faster than functionally identical deployments without it[1]	No significant difference controlling for clinical/functional features
H5	Sovereign Ω, computed from public amendment corpora, leads V-Dem-classified autocratization episode onsets	Median lead time ≤ 0 against permutation null across the episode sample[25]
H6	Procurements subjected to a Trilemma Audit exhibit lower rates of post-deployment conflict-resolution failure than matched unaudited procurements	No significant difference in matched-pair comparison
H7	Auditability A and trailing Ω carry a measurable financing premium (the LACC wedge) in sovereign and institutional funding terms	No significant wedge controlling for macro fundamentals
H8	ZK-attestation of authority reduces the all-in cost of cross-border compliance verification relative to data-transfer-agreement baselines[1]	Proof cost/latency exceeds transfer-agreement cost in the pilot

5.2 Corpora and access

Table 6. Retrodictive and prospective corpora — selection criterion: versioned rule histories with timestamps, predating this framework

Corpus	What it provides	Access
Comparative Constitutions Project[12]	Amendment histories, 200+ national constitutions, 1789–present; documented failure events	Public dataset
HL7 FHIR release & churn history[11]	Profile version churn; breaking-change frequency across DSTU2 → R4 → R5	Public repositories

IETF RFC obsolescence chains[1]	Version flips with procedural metadata at standards-body scale	Public datatracker
Cross-border health exchange incident reports (eHealthDSI / MyHealth@EU-class; national HIE post-mortems)[1]	Failure-mode taxonomy for H2/H3; silent-corruption case material	Public reports; some FOIA-class
V-Dem episode data (ERT)[25],[26]	Independently coded autocratization/democratization episode onsets — the outcome series for H5, never used to tune the instrument	Public dataset
Official gazettes & statutory instruments (pilot jurisdictions)	Sub-constitutional amendment tiers; cooling-off computability at statutory/regulatory level	Public registers
Live CLAI-OS deployment ledgers[1]	Prospective Ω telemetry produced as a schema byproduct — the framework's own reflexivity test bed	Founder-controlled

5.3 Method, pre-registration and null models

For each corpus the method follows the parent work's four-step protocol: (1) reconstruct the amendment ledger as an event series with timestamps, tier classifications and delay compliance; (2) compute φ(t), κ(t), Ω(t) on a rolling window; (3) align against independently documented failure events from the historical literature — *without tuning windows to outcomes*; (4) test the lead-time distribution against a permutation null of 10,000 shuffles of event times.[1] For H2/H3, participating jurisdictions' policy documents are mapped into the pullback formalism and failure clustering is tested against pullback density. For H4 and H6, matched-pair designs hold function constant while auditability varies. For H7, the LACC wedge is estimated on sovereign and institutional funding terms with macro fundamentals as controls, instrumented deployments matched to uninstrumented twins. Pre-registration of window sizes, null models and rejection thresholds occurs before touching outcome data — the framework's own reflexivity discipline forbids tuning the instrument to the result.[1]

5.4 Threats to validity and reflexivity discipline

The programme carries four named threats, each with a mitigation. **Construct validity:** κ is computable only where procedural rules are documented; constitutions with unwritten conventions bias κ downward — mitigated by sensitivity analyses excluding low-documentation systems.[1] **Selection:** failure events are over-documented relative to quiet successes; the permutation null partially corrects but does not eliminate.[1] **Outcome coding:** retrospective classification of failure events carries judgment risk; corpora encode formal amendments, not informal capture — mitigated by using only independently coded outcome series (V-Dem ERT) for H5.[25] **Reflexivity:** if the framework is adopted, its predictions alter the systems it measures; the programme therefore adopts the parent work's own disclaimer — results cease to be naive-valid for systems instrumented by this method after publication — and treats live CLAI-OS ledgers as the dedicated prospective arm where reflexivity is itself the object of study.[1]

> *Primary research question. Can the legitimacy of institutions be metered from their own event ledgers with sufficient lead time, reliability and cheapness that a funder can price it, a regulator can require it, and a state can defend it — before the failure, not after?*

PART VI

The Instrument Suite

6.1 LAS: the accounting schema

The Legitimacy Accounting Standard is deliberately boring technology: an open schema — event tables for traversals, verdicts, amendments, adjudications and refusals; hash-chained per the parent work's migration ledger;[7] CHECK-constrained tier friction (7/90/365-day cooling-off, mandatory sunsets, entrenchment rejected at INSERT) so that capture is "not prohibited by policy but unrepresentable in the data model";[1] and a closed, versioned verdict lexicon including NON_JUSTICIABLE. The standard is published openly because standards create markets: the FHIR playbook — conformance profiles, not products — is the adoption path the parent work identifies, and the same logic applies here.[1] Revenue attaches not to the schema but to the instruments that read it at scale.

6.2 GEWI: the capture early-warning dashboard

The Governance Early-Warning Instrument computes φ, κ, Ω on rolling windows over the Table 6 corpora, maintains pre-registered thresholds per institution class, and issues excursion alerts with full provenance: every alert cites the ledger rows that produced it. For funders, GEWI runs at portfolio level — a single pane over every member country's amendment telemetry, refreshed as gazettes publish, at a marginal cost approaching zero per additional country once the ingestion pipeline exists. The product's integrity guarantee is inherited: because the underlying ledgers are hash-chained, an alert cannot be retroactively erased without visible surgery.[1],[7] GEWI does not predict politics; it reads procedure. That limitation is its credibility.

6.3 The Trilemma Audit protocol

The audit is a fixed-protocol engagement with five stages. **Stage 1 — conflict census:** enumerate the system's conflict classes from its data model and operating procedures. **Stage 2 — pullback mapping:** for each class, pull governing policies back over jurisdiction, consent scope and key version; classify pullback as empty or non-empty per Definition 4.[1] **Stage 3 — regime identification:** for empty-pullback classes, trace the deployed resolution path and identify the sacrificed vertex per relation (3). **Stage 4 — declaration scoring:** score the regime for explicitness (was the sacrifice chosen or stumbled into?), contestability (can losers challenge?), and reversibility (can the regime be renegotiated under versioned amendment?). **Stage 5 — filing:** the audit report enters the procurement file as the declared DPT regime, with the verdict lexicon and complaint schema as the remediation reference. The deliverable is a one-page regime declaration plus the evidence pack — sized to procurement timelines, priced to procurement budgets.

6.4 The Refusal Registry and ZK federation

The Refusal Registry operationalizes the parent work's refusal-set semantics: a published, versioned specification of the conditions under which a system declines to verdict — its stated Gödel sentences — beginning with the NON_JUSTICIABLE class and the re-entry routing rule (the constitution judging its own amendment goes to a human traversal by construction).[1],[10] The registry makes refusals comparable across systems: two deployments can be evaluated on whether their refusal sets are stated, complete and honored. ZK federation is the frontier instrument: zero-knowledge attestation that an act's authority exceeded zero under policy version ν, without revealing the evidence state — compliance without transfer, the graft that replaces the data-

transfer agreement as the atomic unit of cross-border governance.[1] The pilot pairs the registry with one cross-border health-verification corridor.

6.5 The prospective pilots: two-site design

Table 7. Pilot design — two sites, one discipline

Parameter	Pilot A — national health exchange (LMIC)	Pilot B — procurement-authority ledger (MDB portfolio)
Setting	Offline-tolerant health data exchange in a DHIS2-ecosystem country[24]	Rule-change telemetry for a procurement authority supervising a live MDB-financed portfolio[29]
Instruments deployed	LAS schema; verdict lexicon; Trilemma Audit; refusal registry	GEWI; LAS amendment accounts; quarterly legitimacy statements
Primary hypotheses	H2, H3, H4 (failure clustering; silent corruption; review speed)	H1, H5, H7 (lead time; sovereign Ω; LACC wedge)
Duration / cost	18 months / US$1.4 m (design target)	12 months / US$0.6 m (design target)
Success metric	Declared DPT regime filed; Ω telemetry live; review-duration comparison executed	Lead-time distribution vs. permutation null; first LACC estimate
Kill trigger	Pullback mapping infeasible on real policy documents	No lead time in ≥50% of retrodictive cases (H1 rejection)

6.6 Worked illustration: a national health-exchange audit

Consider a stylized national health exchange, 40 facilities, offline 14 hours per day — a composite of documented deployment conditions.[24] The conflict census finds 11 conflict classes; pullback mapping finds 4 empty (consent withdrawal vs. offline access; jurisdiction clash on cross-border referral records; key-version rollover during partition; duplicate-identity merge). Regime identification finds the deployed system silently auto-merging duplicate identities — D sacrificed without declaration — and halting consent-sensitive sync under partition — P partially sacrificed, undocumented. The audit files a declared regime: P and partial-T retained, D restored via adjudication queue with cited merge-ordering criteria, the duplicate-identity class rerouted from auto-merge to first-class conflict state. Remediation cost is dominated by schema adoption, not hardware — the E2 lesson that auditability is an engineering choice.[1] Post-remediation, the exchange produces Ω telemetry as a byproduct and clears its data-protection impact assessment with the ledger as evidence — the H4 channel. Every number in this paragraph is a design target to be replaced by pilot-metered figures; none is a finding.

PART VII

Risk Register and Kill Criteria

7.1 Risk register

Table 10. Programme risks (I = impact, P = probability; H/M/L)

#	Risk	I	P	Mitigation
1	H1 rejection: Ω shows no lead time in retrodictive test	H	M	Pre-registered rejection accepted publicly; pivot value then rests on Trilemma Audit + conformance lines (E2, E3)
2	Corpus coding judgment risk (failure-event classification)	M	M	Independent outcome series only (V-Dem ERT); double-coding with adjudication protocol[25]
3	Pullback formalism inapplicable to real policy texts	H	M	Stage-gated Pilot A kill trigger; fallback to qualitative regime declaration
4	Political sensitivities: sovereign Ω readings read as accusation	H	H	Publish methods, not country league tables; private alerting; institution-first framing
5	Incumbent index providers absorb the idea	M	M	Open standard with reference implementation; corpus flywheel
6	Regulatory delay beyond Dec 2027 softens conformance demand	M	M	Demand diversified across MDB, procurement and DPI channels[28]
7	ZK attestation cost/latency exceeds transfer baselines	M	M	H8 kill criterion; pilot before productization
8	Key-person concentration (founder is the corpus)	H	L	Documentation programme; institutional succession plan; corpus escrow
9	Ledger tampering by coerced operators	M	L	Hash chains make tampering visible, not impossible — stated limitation; distributed witness copies[1]
10	Reflexivity: instrumented systems game Ω	M	M	Reflexivity protocol (§5.4); prospective arm treats gaming as the object of study
11	Funding gap between Phase 1 and anchor subscriptions	M	M	Grant-first sequencing; falsification gates align fund release with evidence
12	Standard captured or forked by a hostile actor	L	L	Versioned governance of LAS itself — on its own ledger, with stated refusal set

7.2 Kill criteria

Table 11. Kill criteria — pre-committed, falsification-forward

ID	Criterion	Consequence
K1	H1 rejected (no lead time in ≥50% of retrodictive cases)	GEWI repositioned as descriptive telemetry; capture-warning claims withdrawn
K2	H2 and H3 both rejected (failures neither cluster on empty pullbacks nor present silently)	Pullback programme closed; structural claims limited to theorem scope

K3	H7 rejected (no LACC wedge)	Financial-pricing narrative withdrawn; pricing case rebuilt on review-speed and failure-cost channels only
K4	H8 rejected (ZK economics unfavorable)	Federation instrument shelved; registry continues as specification only
K5	Two consecutive pilot-stage gates missed	Programme resets to research-only mode; deployment build-out suspended

7.3 The risk-monitoring dashboard

Risk registers fail when they are annual. The programme maintains a live dashboard keyed to the register — corpus coverage, coding progress, threshold integrity, pilot milestones — reviewed monthly, with each kill criterion assigned a named owner and a pre-agreed trigger protocol. The dashboard doubles as public reporting: every quarterly statement reproduces the register's state and the status of each hypothesis, so that reviewers watch the same instrument the team watches. The discipline is the parent work's own: pre-registered falsification, stated refusals, and a ledger that makes revision visible.[1]

SYNTHESIS

Concluding Assessment

Table 17. The argument on one page

Element	Content
Problem	Trillion-dollar exposures rest on institutions whose legitimacy is measured by annual perception, with a conceded lag[15],[18]
Parent result	Legitimacy is a conserved transition-property; DPT forbids D ∧ P ∧ T; Ω is a falsifiable capture signature[1]
Extension	Five contributions: LAS accounts; Trilemma Audit; sovereign-scale GEWI; refusal-set economics + LACC; ZK attestation
Evidence plan	H1–H8 pre-registered; five corpora; permutation nulls; kill criteria K1–K5; reflexivity protocol
Programme stewardship — open standard, four instruments, three constituencies, staged progression on evidence gates	
Timing	Capture wave at record;[25] DPI wave scaling;[23],[24] Art. 12 binding Dec 2027;[28] indices' ceiling conceded[14]

CLOSING STATEMENT

For a century, development finance has priced money and guessed at legitimacy. The parent work ended the guessing in principle: legitimacy is a conserved quantity, its destruction is ledger-visible, and every governed system's deepest design choice — which of due process, partition tolerance and autonomous termination it sacrifices — is enumerable and auditable.[1] This blueprint ends it in practice: an accounting standard, an early-warning instrument, a procurement interrogation, a refusal registry, and an instrument engineered to survive its own falsification criteria. The corpora are public; the method is computational; the rejection criteria are stated in advance. What remains is execution — and the discipline, unusual in this field, of having said exactly what would make us stop.

A final word on intellectual honesty. This blueprint would be the first to meter legitimacy if its programme confirms its hypotheses — and it is written so that the reader can hold it to that conditional. The strongest claim carried by a pre-registered falsification condition is worth more than a revolution asserted. The revolutionary claim here is not that the thesis must be right; it is that the double payment of unmeasured legitimacy — paid once in capture, again in the premium charged to the uninstrumented — is now a metered object, and metering changes what can be financed, insured, regulated and defended. For those who fund it, the commitment is deeper: funding the meter is funding the first governance instrument that can prove it is working — or prove, on the record and at its own expense, that it is not.

EVIDENCE

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[24] DHIS2 / HISP Centre, University of Oslo. (2026). "About DHIS2" — world's largest HMIS platform; government-wide system of record for health data in 80+ low- and middle-income countries; 100+ countries incl. NGO programmes; designed for limited-connectivity environments.

[25] V-Dem Institute. (2026). *Democracy Report 2026* — 44 countries autocratizing (record); 3.4 bn people (41% of world population) and 39% of world GDP in autocratizing countries; democracy for the average citizen back to 1978 levels; media censorship in 32 of 44 autocratizers (73%). University of Gothenburg, March 2026.

[26] V-Dem Institute. (2025). *Democracy Report 2025: 25 Years of Autocratization* — 45 autocratizers; of 27 episodes beginning in democracies, only 9 remained democracies (67% fatality rate); 72% of world population in autocracies. University of Gothenburg, March 2025.

[27] European Union. (2024). *Regulation (EU) 2024/1689 (AI Act)*, Article 12 — high-risk AI systems "shall technically allow for the automatic recording of events (logs) over the lifetime of the system"; penalties to €15 m or 3% of global annual turnover (Art. 99, tier 2).

[28] European Union. (2026). *Regulation (EU) 2026/1744 (Digital Omnibus on AI)*, in force 27 July 2026 — high-risk obligations (incl. Art. 12 record-keeping) applicable from 2 December 2027 (Annex III) / 2 August 2028 (Annex I); compliance analyses: audit-grade evidence requires 12+ months of production log history.

[29] IBRD. (2025). *Financial Statements, June 2025* — FY2025 net loan commitments US$40.9 billion across 139 operations (+US$3.3 bn / +9% vs FY24); FY2026 SALL-Adj US$42.0 bn.

[30] United Nations. (2024). *Global Digital Compact*, adopted at the Summit of the Future, September 2024; 315 endorsements from entities in 93 countries as of 31 December 2024 (South Centre Policy Brief 140, 2025).

APPENDIX A

Glossary of Constructs

Table A1. Defined terms

Term	Definition
Traversal	Six-phase authorization loop — observe, infer, authorize, act, verify, return — gated by computed authority under the constitutional version in force[1]
Legitimacy of a transition	Three-clause property: authority > 0 at traversal time; hash-chained ledger record; continuing contestability[1]
Eigenform	Fixed point of the verification operation: re-running the traversal on a transition's provenance returns the same verdict[1]
DPT theorem	No governed sync system guarantees due-processed resolution ∧ partition tolerance ∧ autonomous termination[1]
Pullback / pushout	Category-theoretic test for policy common ground; empty pullback ⇒ conflict structurally inevitable ⇒ route to adjudication preemptively[1],[9]
Ω (oscillation index)	$\Omega(t) = \varphi(t)\cdot(1-\kappa(t))$: version-flip frequency times cooling-off non-compliance — candidate capture signature[1]
LAS	Legitimacy Accounting Standard: double-entry discipline posting every transition's legitimacy score to a verifiable ledger (this blueprint, contribution 1)
L(t), U(t), Φ(t)	Legitimacy stock, ultra-vires deficit, and legitimate-transition flow — the accounting objects of relation (1)
GEWI	Governance Early-Warning Instrument: sovereign-scale Ω monitor with pre-registered thresholds (contribution 3)
Trilemma Audit	Five-stage procurement protocol producing a declared, auditable DPT sacrifice regime (contribution 2)
NON_JUSTICIABLE	Verdict class declaring refusal to verdict — the system's stated Gödel sentence; re-entries route to human traversal[1]
LACC	Legitimacy-adjusted cost of capital: $r_i = r_f + \rho(1-A_i) + \pi\cdot\Omega_i(1-A_i)^{1/2}$ (contribution 4)
ZK attestation of authority	Proving authority > 0 under policy version ν without revealing evidence state E — compliance without transfer[1]
Reflexivity discipline	Pre-registration, no tuning to outcomes, and the disclaimer that instrumented systems cease to be naive-valid[1]

APPENDIX B

Indicator and Verification Matrix

Table B1. Programme indicators — every indicator has a source, a cadence and a hypothesis it serves

Indicator	Source instrument	Frequency	Hypothesis served
Ω(t) per institution/country	GEWI over amendment ledgers	Rolling (weekly)	H1, H5
Cooling-off compliance κ(t)	Gazette/ledger tier tagging	Per amendment	H1, H5
Pullback density ρ_PB	Trilemma Audit stage 2	Per engagement	H2
Silent-corruption share of failures	Incident taxonomy coding	Per corpus pass	H3
Regulatory review duration	Matched-pair deployment records	Per deployment	H4
Declared-vs-undeclared regime rate	Procurement file census	Quarterly	H6
LACC wedge estimate	Financing-terms panel, matched twins	Semi-annual	H7
ZK proof cost/latency vs. transfer baseline	Federation pilot telemetry	Continuous in pilot	H8
Legitimacy stock L(t), deficit U(t)	LAS statements	Quarterly	All (context)

APPENDIX C

Capture Case Files

The retrodictive programme codes episodes against the Ω signature. The files below define the coding frame; all substantive episode classifications are drawn from the independently coded V-Dem ERT series,[25],[26] never from the programme's own judgment.

Table C1. Case-file coding frame (episode classes, not verdicts)

Class	Definitional anchor	Ω prediction under H1/H5
Stand-alone autocratization	Episode beginning from democratic or autocratic start with sustained LDI decline (24 of 44 current cases)[25]	Ω excursion precedes episode onset; κ erodes first at constitutional tier
Bell-turn autocratization	Democracy that turns and declines (20 of 44)[25]	Ω leads the turn; procedural tactics (censorship, civil-society rules) visible as flips
U-turn democratization	Reversal of an autocratization episode (subset of 18 democratizers)[25]	Ω re-bounds: flip frequency falls, κ recovers — the signature in reverse
Stand-alone democratization	Deepening from democracy or transition from autocracy (8 ongoing; combined population ~40 m)[25]	Bounded Ω throughout — the healthy-band control class
Standards-body capture	Vendor/interest capture of specification process (FHIR/IETF corpora)[1],[11]	Churn signature: breaking-change frequency rises with eroded review discipline

APPENDIX D

Sensitivity Analysis

The thesis's exposure to its four most volatile parameters is bounded below. The stress envelope differs from the base case principally in timing: GEWI lead time, audit-conversion rate, LACC wedge size and pilot funding each shift the adoption curve, not its direction — because the empirical programme's cost is small relative to the procurement and conformance markets it addresses.

Table D1. Wedge sensitivity (base case; illustrative)

Parameter shock	Effect on GEWI	Effect on audits	Effect on LACC case
Lead time −50%	Alert value falls; descriptive telemetry remains	None	Wedge narrative survives on review-speed channel (H4)
H7 rejected (no wedge)	None	None	Pricing claims withdrawn (K3); pricing case rebuilt on H4 + failure-cost channel
Regulatory delay to 2028+	None	Conformance demand shifts right[28]	None
Corpus documentation gaps	κ biased down in low-documentation systems; sensitivity exclusion applies[1]	Pullback coding slower	None
Pilot budget −40%	Retrodictive core unaffected (public data)	Pilot A scope reduced to ledger + audit	H7 estimation deferred to Phase 3

The directional conclusion is robust: no single observed-variable shock closes the framework's case, because the four timing forces of Part I are independent. What would close it is stated in Table 11 — which is why the programme stages its claims behind falsification gates and treats a clean rejection as a publishable, valuable result.[1]

> **Endnote on evidence discipline.** All quantitative claims in this blueprint trace to the cited sources as of October 2026. Where estimates differ between sources (e.g., corruption-cost magnitudes[13],[14]), the range is stated and the methodological caveat is quoted. Forward-looking quantities — pilot budgets, lead times, wedge sizes — …are design targets to be replaced by metered figures under the pre-registered protocols of Part V, not findings. This public edition is a research blueprint; it does not constitute investment advice, a legal opinion, or a guarantee of empirical outcomes. The falsification gates of Table 11 are the programme's own commitment: a clean rejection is a publishable result, and the blueprint's claims stand only until its instruments say otherwise.

Appendix H-H — Reconciliation Note: GEWI as Early-Warning Instrument (note to §3.3)

Reconciliation note (instrument supersession). The Ω-threshold alerting instrument described in this section is the first-generation construct, and it has since been superseded by registered amendment. White Paper Nº 3 (The Registered Frontier, §2.2 and amendments A1/A2) replaced the composite Ω trigger with a two-channel detector telemetry — version-flip frequency ρ and cooling-off compliance λ, scored by cumulative-sum accumulation with a conjoint alarm condition — after the retrodictive programme showed that the single composite conflated regimes the two channels separate (the reformer pattern $\rho\uparrow$ alone; the capture pattern $\rho\uparrow \wedge \lambda\downarrow$). Notation is unified across the series as follows. φ (amendment frequency whose effect invalidates acts authorized under the amended rules) and κ (cooling-off compliance) remain the ledger-level constructs, defined here and imported unchanged by the parent work's descendants; ρ and λ are the corresponding detector telemetry, computed on rolling windows over public corpora. Ω remains in use only as the trailing capture-signature exposure entering the legitimacy-adjusted cost of capital (§3.4); it is no longer an alerting instrument. Readers implementing against this blueprint should implement the Nº 3 telemetry; this section is retained as the motivation and the retrodictive design history.